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India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

10 MILLION SOLAR ROOFTOP SYSTEMS AND MANAGEMENT OF REVERSE POWER FLOWS ON DISTRIBUTION FEEDERS

Presented By

Brajesh Kumar-COO BSES Yamuna Power Limited

Discom Profile

BSES Yamuna Power Ltd

51 : 49 Joint Venture
of R-Infra & GoNCTD formed on 1st July 2002

- ✓ *Successfully serving the **National Capital for over two decades…***
- ✓ BYPL supplying power to **historic, theft prone and most challenging high density** old city areas of Delhi with consumer density of ~12,700/sq km and low HT to LT Network ratio of 0.56
- ✓ *Caters **22% of National Capital's demand**, spread across EAST & parts of CENTRAL Delhi*
- ✓ One of the most Successful Public Private Partnership. *An **ISO 9001:2015, ISO 14001:2015 & BS OHSAS 45001:2018 Certified** Organization*

Key Parameters	FY25
Distribution Area	160 sq KM
Consumers	**2.04 million**
Peak Load	**1,882 MW** (on 19th Jun-24)
Customer Density	**~12,700 /Sq.Km**
T&D Loss	**6.57%**
Reliability Index	**99.9%**
EHV Capacity	4,046 MVA
Distribution Capacity	3,801 MVA
HT Cable Length	3,467 ckt km
LT Cable Length	6,171 ckt km

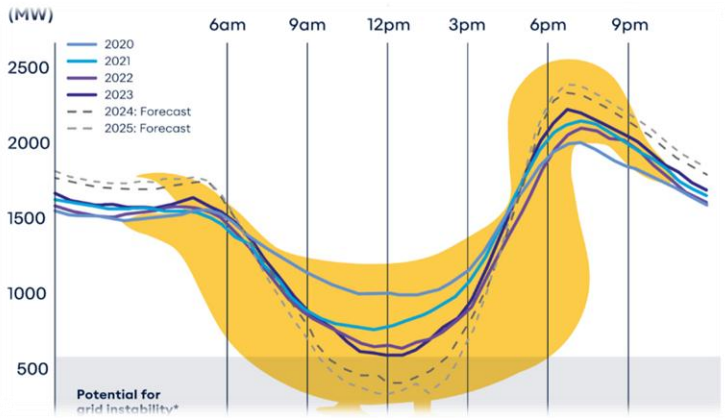
Towards progressive and sustainable journey for improving customer services & deployment of cutting edge technologies

Context

- PM Surya Ghar scheme targeting 10 million rooftop solar systems (~30 GW).
- Rapid RTS growth reshaping urban distribution networks.
- Urban DISCOMs could be facing reverse flow, voltage and operational challenges.
- Grid readiness is now critical for rooftop solar success.

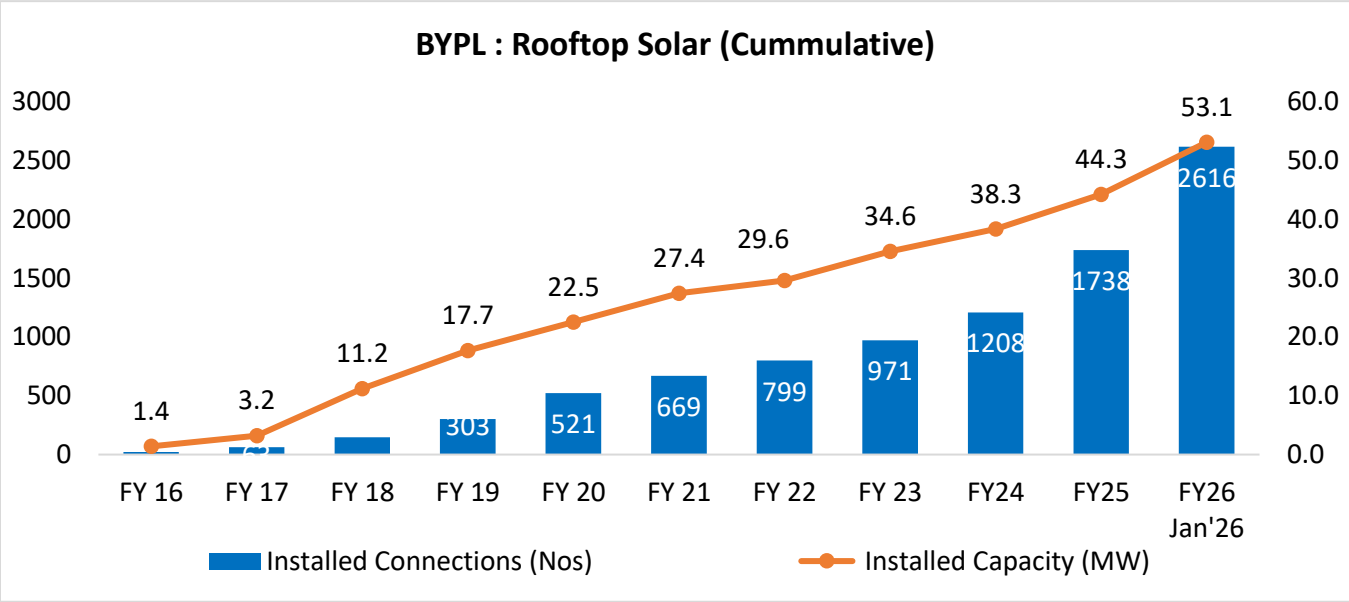
Emerging Grid Challenges

- Voltage rise and inverter tripping during solar peak.
- Reverse power flow in dense urban feeders.
- Deepening duck curve and forecasting uncertainty.
- Commercial and tariff pressures on DISCOMs.



Relevence

- Growing rooftop solar penetration across consumer segments.
- High smart meter and digital system adoption underway.
- Focus on reliable, technology-driven network operations
- With progressive regulations Installed capacity has shown significant rise over the years
- Target is to add 20 MW from rooftop PV by FY 27
- BYPL in the process of aggregating capacities under ULA model to target 10,000 installations



BYPL RTS installed capacity is 53.1 MW & RTS Installations are 2616 nos (Jan 26)

Smart Inverters: First Line of Defense

- Volt-VAR and Volt-Watt capabilities mitigate voltage excursions.
  - When reverse power causes voltage to rise, the inverter absorbs reactive power (VAR), reducing the voltage to safe levels.
  - If the grid voltage becomes too high (e.g., above 110% of nominal value), the inverter automatically reduces its active power (Watt) output, preventing overvoltage without needing to disconnect

- Ride-through and remote configurability improve reliability.
- If required smart inverters can reduce active power fed back to the grid
- Interoperability and utility visibility are critical.

Unlike traditional, passive inverters, smart inverters use advanced control algorithms to prevent overvoltage, equipment damage, and instability in distribution networks.

BESS Deployment Strategies

- Consumer-level storage to increase self-consumption.
- DT-level BESS for local voltage and reverse flow management.
- Substation BESS for system flexibility and peak support.
- Data-driven, targeted deployment preferred over blanket rollout .
- Decentralized small installations better as compared to large Grid level interventions owing to very low space availability for DT augmentation/addition (specific to discom)
- Reverse power flow is often accompanied by high voltage levels in distribution networks. BESS inverters can manage reactive power (VARs) to help stabilize voltage levels, preventing overvoltage conditions that occur when excessive power is pushed onto the grid
- **Zero Export Solutions:** BESS can be configured to act as a "zero-export" system, ensuring all generated solar power is either consumed on-site or stored in the battery, completely eliminating export to the grid

EV Charging as Flexible Load

- Daytime EV charging can absorb excess solar generation.
- Smart charging enables dynamic load balancing
  - If an energy management system (EMS) detects that a local transformer is experiencing reverse power flow, it can command EVs to increase their load, bringing the net power flow back towards zero.
- Charging can be aligned with rooftop generation forecast
- Utilities need to identify solar-rich feeders for planning smart EV infrastructure.
- V2G and V2H : EVs as distributed storage that mitigates voltage volatility caused by reverse power flow

Demand-Side Electrification

- Electric cooking can create useful daytime demand.
- District Cooling Systems provide large urban daytime loads.
- Electrification must complement solar growth.
- Appliance ecosystem and incentives are essential.

Dynamic Tariff Signals

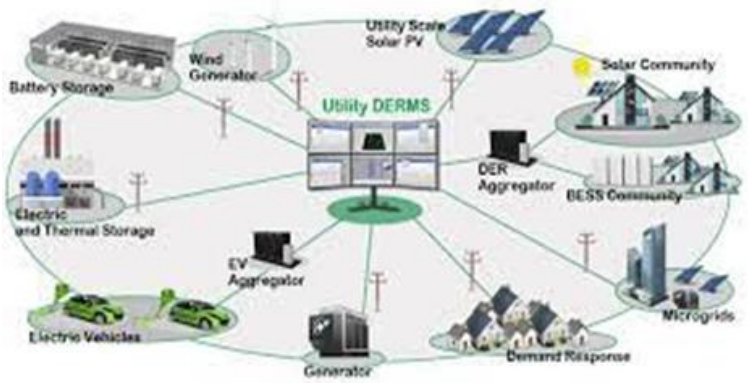
- Solar-hour concessional tariffs to flatten duck curve.
- Dynamic ToD and critical peak pricing mechanisms.
- Encourage consumption during high solar generation hours.
- Tariff design must remain simple and predictable.

DERMS: The Control Layer

- **Future grid will manage millions of DER assets.**
- **Real-time monitoring, forecasting and active control required.**
- **Integration with ADMS and feeder visibility critical.**
- **Pilot-to-scale roadmap underway.**

An **application platform** to manage device information, **monitor and enable optimization and control of DER* and DR**.** A DERMS must be able to aggregate, simplify, optimize, and translate DER and DR functionalities..

- A platform or system-of-systems
- Provides visibility of DERs and ability to optimally control DERs connected to the electric distribution system
- Used by system operators, planners, and other utility stakeholders



DER Monitoring and Control:

- Near real-time monitoring of status and output of individual DERs
- Control of individual DERs (On/Off control, setpoint control, etc.)

Forecasting and Estimation

- Weather and other parameter-adjusted forecasting of DR and DER assets (short term, medium term, longer term)
- Estimation of near-real-time output for non-metered DER assets

Modeling of the interconnected DER assets

- Supply DER model info required by ADMS and system planning
- Nameplate data (rating, impedance,…)
- Location
- Contracted operational constraints

Recommended Policy Actions

- Adherence to National smart inverter and communication standards.
- Storage and flexible demand incentives required.
    - Demand Flexibility regulations-State specific
- Feeder-level hosting capacity maps.
- Urban DER planning framework and potential mapping

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Links/References (If any)